

SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26135
- **Problem Statement Title-** Difficulties in tracking employment outcomes, skill gaps, and the impact of skilling initiatives
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)-** Code Storm



Problem Understanding

☐ Who is facing the problem?

Government agencies, training institutes, job seekers, and employers are affected by this problem.

☐ What problem are they facing?

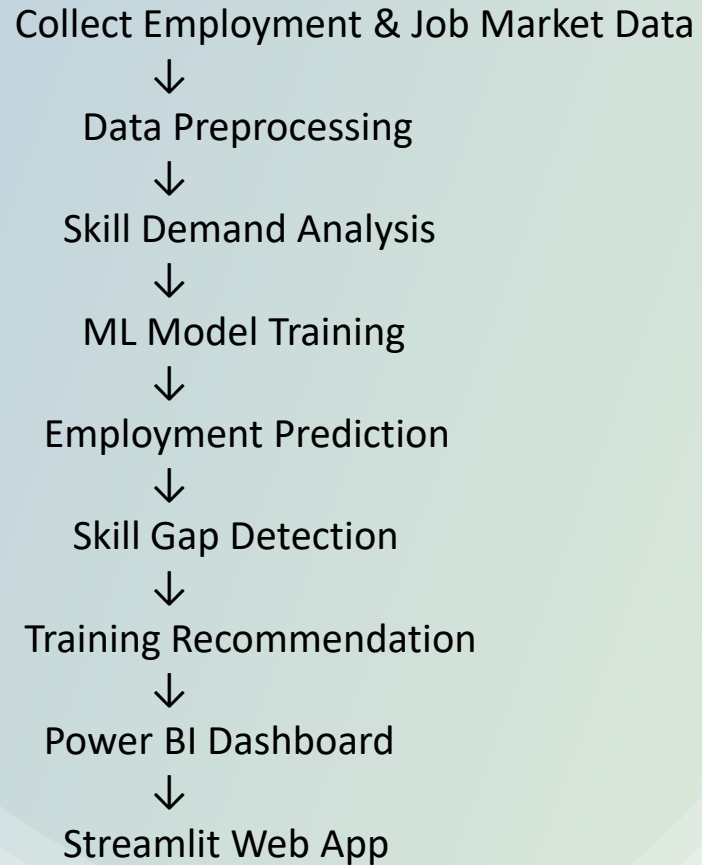
They face difficulties in tracking employment outcomes, identifying skill gaps, and measuring the effectiveness of skilling programs.

☐ Why are they facing the problem?

This is mainly because employment, training, and skill data are scattered across different sources and are difficult to analyze together.

Tentative Technologies: Python, Pandas, NumPy and Scikit-learn will be used for data analysis and ML, while Power BI will be used for visualization and Streamlit for the web interface.

Methodology:



Feasibility And Viability

❑ Feasibility:

The solution is technically and economically feasible using Python, Machine Learning, Power BI, and Streamlit.
Existing employment and job-market datasets can be used for development and testing.

❑ Challenges & Risks:

Employment data may be incomplete or inconsistent.
Changing industry requirements may affect skill-gap analysis.
ML predictions may not always be accurate.

❑ Strategies:

Clean and standardize data before analysis.
Regularly update job-market and skill data.
Evaluate and improve the ML model using suitable performance metrics.

Impact And Benefits

❑ Potential Impact:

The solution can help government agencies, training institutes, job seekers, and employers make better data-driven decisions.

It can improve employment tracking, skill-gap identification, and training effectiveness.

❑ Benefits:

Social: Helps job seekers find relevant skills and training opportunities.

Economic: Improves employability and supports better workforce planning.

Institutional: Helps training providers evaluate and improve their programs.

Overall: Promotes better alignment between industry skill requirements and workforce skills

References & Research Links

AI-Powered Resume Screening Dataset – Used for the ML-based resume screening model.

link: kaggle.com/datasets/mdtalhask/ai-powered-resume-screening-dataset-2025

Indian Job Market Dataset – Used for analysing in-demand skills, job roles, salaries, and employment trends.

link: kaggle.com/datasets/shivamshrivastava21/indian-job-market-dataset-2025-2026

Python, Pandas & Scikit-learn – Used for data preprocessing, skill analysis, and ML prediction.

Power BI & Streamlit – Used to visualize insights and develop the prototype interface.